Spring 22

1) a) My ID is Z233040, so x is 0

i) The base criteria of $G(N)$ is when $N<5$, and in this case:

$G(N) = 3 \times N$

ii) $G(2)$ ~~= 3~~

$N = 2$

$G(2) = 3 \times 2 = 6$

$G(8)$

$N = 8$

~~$2 \times N(8) =$~~ $G(N) = 2 \times G(N-5) + x$

$G(8) = 2 \times G(8-5) + 0$

$= 2 \times G(3)$

$G(3) = 3 \times 3 = 9$

$\therefore G(8) = 2 \times 9 = 18$

$G(24)$

$N = 24$

$G(24) = 2 \times G(24-5) + 0$

$= 2 \times G(19)$

$G(19) = 3 \times 19$

$= 57$

$G(24) = 2 \times 57$

$= 114$

1) b) FRONT = 2, REAR = 5

i) ~~E~~ E, —, U, C, S, — FRONT = 2, REAR = 0

ii) E, —, —, —, S, — FRONT = 4, REAR = 0

iii) E, U, V, A, S, — FRONT = 4, REAR = 3

iv) —, —, V, A, —, — FRONT = 1, REAR = 3

c) ~~Q(~~
QINSERT (QUEUE, N, FRONT, REAR, ITEM)

Step 1: [QUEUE already filled?]
If FRONT = 1 and REAR = N, or ~~if~~
if FRONT = REAR + 1, then:
Write: OVERFLOW and return

Step 2: [Find new value value of REAR]
If FRONT = NULL, then:
Set FRONT := 1 and REAR := 1
Else if REAR = N, then:
~~R~~ Set REAR := 1
Else
REAR := REAR + 1
[End of If structure]

~~Step 3:~~

Step 3: Set QUEUE[REAR]:= ITEM
[This inserts a new element]

Step 4: Return

QDELETE(QUEUE, N, FRONT, REAR, ITEM)

Step 1: [QUEUE already empty?]
If FRONT = NULL, then:
Write: UNDERFLOW and return

Step 2: Set ITEM := QUEUE[FRONT]

Step 3: [Find new value of FRONT]
If FRONT = REAR, then:
Set FRONT := NULL and REAR := NULL
Else if FRONT = N, then:
~~S~~ Set FRONT := 1

Else
Set FRONT := FRONT+1
[End of if structure]

Step 4: Return.

(Group B)

~~A~~ Answer to the question No. 3

~~a)~~ 3)

a) Hash function: A hash function is any function that can be used to map data of arbitary size to fixed-sized values. The values returned by a hash function are called hash ~~values~~ or values or hash codes. The values are used to index a fixed ~~value~~ size table called a hash table.

b) i) The complexity of linear search algorithm:

It takes up no ~~space~~ extra space; its space complexity is O(n) for an array of ~~a~~ n elements

ii) The complexity of bubble sort algorithm:

It is a reliable sorting algorithm. This algorithm has a worst-case time complexity of $O(n^2)$. The bubble sort has a space complexity of $O(1)$.

iii) The complexity of Binary search Algorithm: $O(\log n)$

The best-case time complexity would be $O(1)$ when the central index directly matches with the desired value.

iv) <u>The complexity of Insertion Sort Algorithm:</u>
$O(n^2)$. The best-case time complexity of insertion sort algorithm is $O(n)$.

v) <u>The complexity of selection sort Algorithm:</u>
$O(n^2)$. selection sort Algorithm has a space complexity of $O(1)$ because it takes some extra memory space for temp variable for swapping.

(C)

Pass	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]
K=1, LOC=2	B	A	N	G	L	A	D	E	S	H
K=2, LOC=6	A	B	N	G	L	A	D	E	S	H
K=3, LOC=6	A	A	N	G	L	B	D	E	S	H
K=4, LOC=7	A	A	B	G	L	N	D	E	S	H
K=5, LOC=8	A	A	B	D	L	N	G	E	S	H
K=6, LOC=7	A	A	B	D	E	N	G	L	S	H
K=7, LOC=10	A	A	B	D	E	G	N	L	S	H
K=8, LOC=8	A	A	B	D	E	G	H	L	S	N
K=9, LOC=10	A	A	B	D	E	G	H	L	S	N
~~K=10, LOC=~~										
sorted:	A	A	B	D	E	G	H	L	N	S

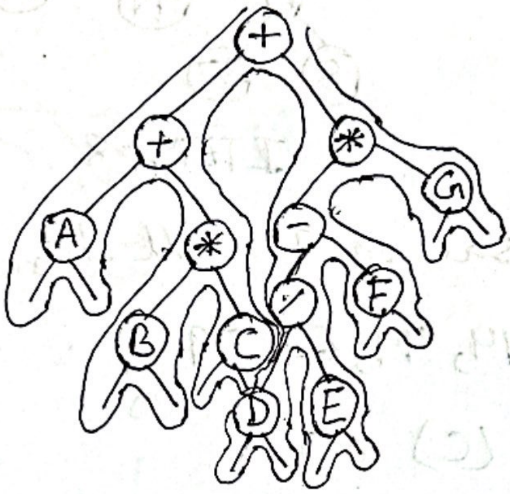
(C) (or)

~~Pass~~	~~A[0]~~	~~A[1]~~	~~A[2]~~	~~A[3]~~	~~A[4]~~	~~A[5]~~	~~A[6]~~	~~A[7]~~							
Pass	A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]	A[12]	A[13]	A[14]
K=1	$-\infty$	D	A	T	A	S	T	R	U	C	T	U	R	E	S
K=2	$-\infty$	D	A	T	A	S	T	R	U	C	T	U	R	E	S
K=3	$-\infty$	A	D	T	A	S	T	R	U	C	T	U	R	E	S
K=4	$-\infty$	A	D	T	A	S	T	R	U	C	T	U	R	E	S
K=5	$-\infty$	A	A	D	T	S	T	R	U	C	T	U	R	E	S
K=6	$-\infty$	A	A	D	S	T	T	R	U	C	T	U	R	E	S
K=7	$-\infty$	A	A	D	S	T	T	R	U	C	T	U	R	E	S
K=8	$-\infty$	A	A	D	R	S	T	T	U	C	T	U	R	E	S
K=9	$-\infty$	A	A	D	R	S	T	T	U	C	T	U	R	E	S
K=10	$-\infty$	A	A	C	D	R	S	T	T	U	T	U	R	E	S
K=11	$-\infty$	A	A	C	D	R	S	T	T	T	U	U	R	E	S
K=12	$-\infty$	A	A	C	D	R	S	T	T	T	U	U	R	E	S
K=13	$-\infty$	A	A	C	D	R	R	S	T	T	T	U	U	E	S
K=14	$-\infty$	A	A	C	D	E	R	R	S	T	T	T	U	U	S
Sorted:	$-\infty$	A	A	C	D	E	R	R	S	S	T	T	T	U	U

Answer to the question No. 4.

(a)

(A+B*C) + ((D/E) - F) * G)



Preorder: + + A * B C * - / D E F G

Inorder: A + B * C + D / E - F * G

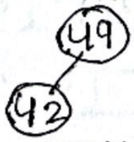
Postorder: A B C * + D E / F - G * +

(c)

49, 42, 19, 62, 37, 79, 44, 8

i)

ITEM = 49



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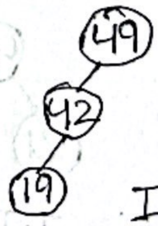

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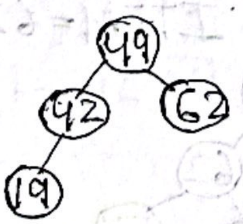
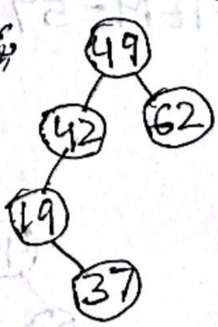
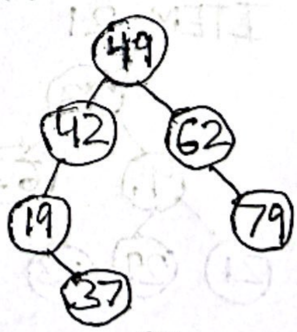


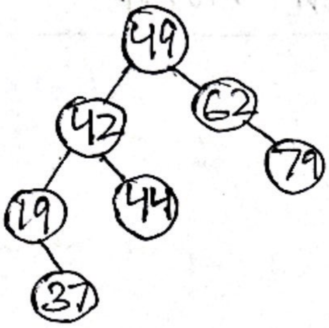
ITEM = 62



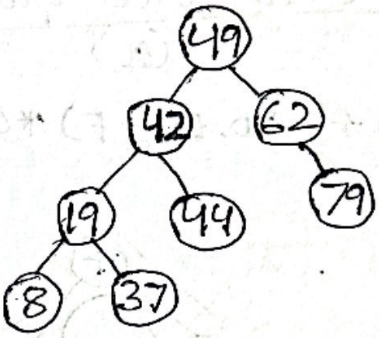
ITEM = 37



ITEM = 79



ITEM=44



ITEM=8

ii) Inorder traversal of T will be,

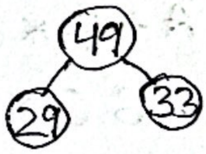
8, 19, 37, 42, 44, 49, 62, 79

(c) (or)

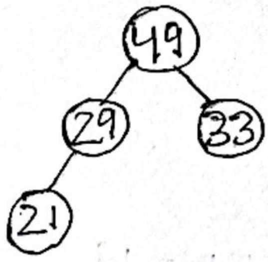
33, 29, 49, 21, 57, 62, 73, 54

ITEM=33

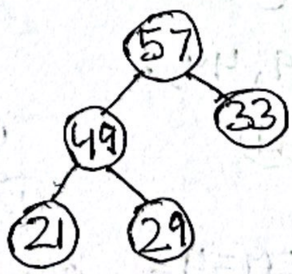
ITEM=29



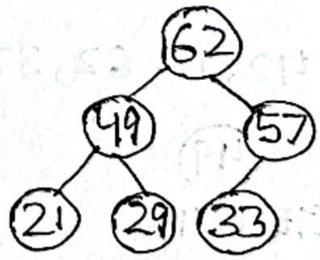
ITEM=49



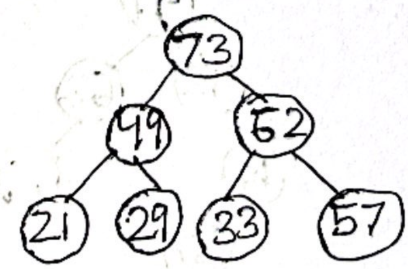
ITEM=21



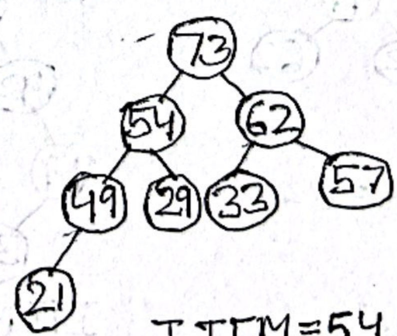
ITEM=57



ITEM=62



ITEM=73

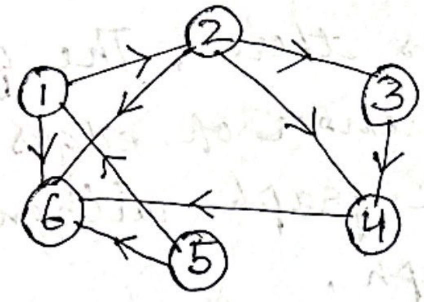


ITEM=54

Answer to the question No. 5

(a)

i)



i) Adjacency matrix :

$$\begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 2 & 0 & 0 & 1 & 1 & 0 & 1 \\ 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 & 0 & 1 \\ 5 & 1 & 0 & 0 & 0 & 0 & 1 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

ii) Adjacency list

1 → 2 → 6 X

2 → 3 → 4 → 6 X

3 → 4 X

4 → 6 X

5 → 1 → 6 X

6 X

(b)

For a set of vertices V with n elements n(n-1)/2 possible edges there. The graph containing maximum number of edges in a n element undirected graph without self loop is complete graph.

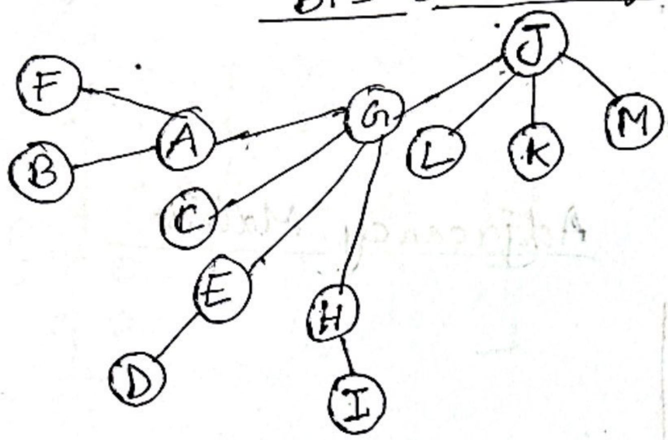
(c)

BFS	DFS
1) BFS, stands for Breadth first search.	1) DFS, stands for Depth first search.
2) It ~~used~~ uses Queue to find the shortest path	2) It uses stack to find the shortest path.
3) Slower than DFS	3) Faster than ~~DFS~~ BFS

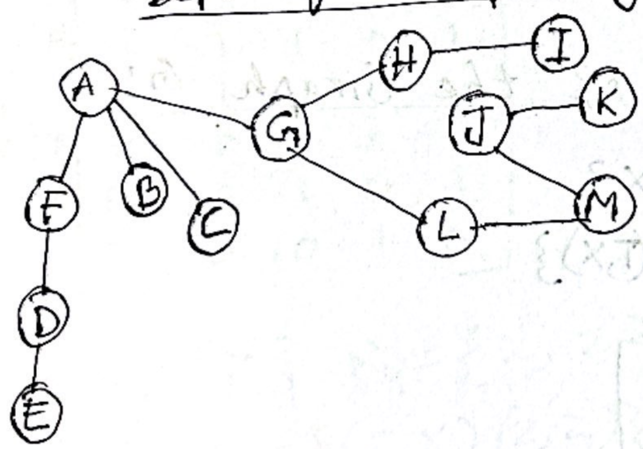
(d)

My ID is C233040. (even)

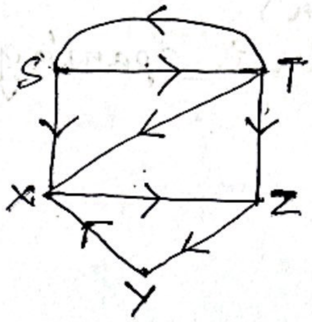
BFS Breadth first spanning tree



DFS
Depth first spanning tree



Answer to the question No. 5(d)(or)



Adjacency Matrix

i)

$$P_0 = \begin{array}{c} \\ S \\ T \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} S & T & X & Y & Z \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

ii) **The path matrix P of the Graph G:**

$C_1 = \{T\}$ $R_1 = \{T, X\}$

$C_1 \times R_1 = \{(T, \cancel{X}T), (T, X)\}$

$\cancel{P_x}$

$$P_1 = \begin{array}{c} \\ S \\ T \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} S & T & X & Y & Z \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$C_2 = \{S, T\}$ $R_2 = \{S, T, X, Z\}$

$C_2 \times R_2 = \{(S,S), (S,T), (S,X), (S,Z), (T,S), (T,T), (T,X), (T,Z)\}$

$$P_2 = \begin{array}{c} \\ S \\ T \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} S & T & X & Y & Z \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$C_3 = \{S, T, Y\}$ $R_3 = \{Z\}$

$C_3 \times R_3 = \{(S, Z), (T, Z), (Y, Z)\}$

$$P_3 = \begin{array}{c} \\ S \\ T \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} S & T & X & Y & Z \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$C_4 = \{Z\}$ $R_4 = \{X, Z\}$

$C_4 \times R_4 = \{(Z, X), (Z, Z)\}$

$$P_4 = \begin{array}{c} \\ S \\ T \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} S & T & X & Y & Z \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

$C_5 = \{S, T, X, Y, Z\}$ $R_5 = \{X, Y, Z\}$

$C_5 \times R_5 = \{$

$$P_5 = \begin{array}{c|ccccc} & S & T & X & Y & Z \\ \hline S & 1 & 1 & 1 & 1 & 1 \\ T & 1 & 1 & 1 & 1 & 1 \\ X & 0 & 0 & 1 & 1 & 1 \\ Y & 0 & 0 & 1 & 1 & 1 \\ Z & 0 & 0 & 1 & 1 & 1 \end{array}$$